

Linea Burn Mechanism Audit Summary

Prepared by Cyfrin

Version 2.2

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Linea on Linea Burn Mechanism. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.2 dated November 3, 2025, was delivered to Linea, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Linea Burn Mechanism code provided by Linea. The findings consist of 1 Low severity issue with the remainder being informational and gas optimizations.

Summary

Project Name	Linea Burn Mechanism
Repository	contract-freeze-2025-10-12
Commit	8285efababe0...
Fix Commit	77b83d2ce990...
Audit Timeline	October 20th - October 24th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	1	1	0
Informational	10	7	3
Gas Optimizations	2	2	0
Total	13	10	3

6 Audit Scope

The audit scope was at commit 8285efababe0689aec5f0a21a28212d9d22df22e:

```
contracts/src/operational/L1LineaTokenBurner.sol
contracts/src/operational/RollupRevenueVault.sol
contracts/src/operational/V3DexSwap.sol
contracts/src/operational/interfaces/IL1LineaTokenBurner.sol
contracts/src/operational/interfaces/IRollupRevenueVault.sol
contracts/src/operational/interfaces/IV3DexSwap.sol
```

7 Deployment Verification

Post-audit fixes. Fixes were applied in commit 77b83d2. A minor NatSpec adjustment to rename a variable was applied in commit efe83ff; this change does not affect the bytecode or ABI of the contracts. The bytecode and ABI from the `main` branch have been verified to exactly match the versions following the post-audit fixes. Both commits have been merged into the `main` branch of `linea-monorepo` and are included in the release tag `contract-audit-2025-10-30`.

Deployed contracts. The following contracts have been deployed:

- **RollupRevenueVault** on Linea: `0x84a5ba2c12a15071660b0682b59e665dc2faaedb`

- **V3DexSwapAdapter** on Linea: 0x30a20a3a9991c939290f4329cb52daac8e97f353
- **L1LineaTokenBurner** on Ethereum Mainnet: 0x5Ad9369254F29b724d98F6ce98Cb7bAD729969F3

The deployed bytecode of all three contracts has been verified and confirmed to exactly match the corresponding bytecode generated from the source code at the release tag `contract-audit-2025-10-30`.

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

The Linea Burn Mechanism implements a dual-token burn protocol designed to allocate Layer 2 (L2) transaction fees toward operational reimbursement while simultaneously enhancing the strength of Ethereum (`ETH`) and the Linea (`LINEA`) token.

- **Operational Reimbursement**: Network operators submit invoices for reimbursement, with payments drawn directly from the L2 fee revenue
- **Surplus Allocation and Token Burn**: Following full settlement of operational expenditures, any residual fee surplus is deployed to execute market purchases of `LINEA` tokens. These acquired tokens are subsequently removed from circulation through permanent burning, thereby reducing the `LINEA` token supply

10 Attestation

Cyfrin attests that it performed a security review of the Linea Burn Mechanism code provided by Linea.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-11-03-cyfrin-linea-burn-v2.2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Linea has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=24114cd42e41b630c2a986dd437a3b982289d96e26ece62a3dc25809ef8c18d7
F=2025-11-03-cyfrin-linea-burn-v2.2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.